

Endpoint-matched monitoring prevents overclaiming

a Endpoint-matched monitoring workflow

Risk question	Shock or gate	Observable endpoint	Allowed claim	Production action
Where may exposure matter?	▶ Structural source-country dependence	▶ UN Comtrade source-country and HS6 exposure	▶ Exposure pathway, not disruption	▶ Supplier watchlist and targeted evidence collection
Can exposure become disruption?	▶ Mine, route, port, licence, conflict or logistics gate	▶ Official production, route, port or administrative endpoint	▶ Gate-conditioned disruption hypothesis	▶ Inventory review and alternate-source readiness
Is an exchange-rule shock observable?	▶ Warranting, origin reporting, brand eligibility or exchange rule	▶ Origin labels, warrant categories and off-warrant eligibility	▶ Deliverability reclassification	▶ Eligible-stock monitoring and contract review
Did it become broad market response?	▶ Market-context or price-curve response	▶ Regional premiums, Cash-3M spreads and price windows	▶ Market impact only if nulls are defeated	▶ Hedging or contract adjustment only after null defeat

b Current evidence status and claim ceiling

Evidence layer	Status	Support	Claim ceiling
Global exposure baseline	supported	Tin top-1 source share 27.8%; nickel top-5 source share 51.3%.	Maps exposure pathways only.
LME deliverability endpoint	supported	Origin endpoint AUC 1.000; exact p 0.0079.	Exchange-rule observability, not broad market impact.
Price and premium endpoints	boundary null	Cash-3M minimum BH q 0.0685; aluminium premium null rows 6.	Broad price transmission is unsupported.
Indonesia tin source closure	ready boundary	Jan-Feb HS8001 zero tonnes; March 3802.5 tonnes; near-zero run 2/2.	Trade-gate consistency check, not permit-level causality.
Permit-level tin causality	blocked	Company-month admin rows recovered: 0.	Cannot claim RKAB/MOMS/SIMBARA causal timing.
Universal predictive risk score	blocked	Evidence is endpoint-specific and general validation remains bounded.	Use framework, not a universal predictor.

Source: derived demonstration tables only. Green marks supported endpoint evidence; yellow marks boundary/null evidence; red marks blocked claims.